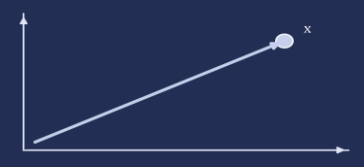
Machine Learning Foundations: Geometry, Optimization, and Linear Classification

From vectors and planes to loss functions, feature vectors, and classifiers

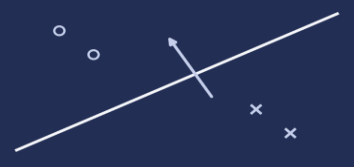
Dr. Yael Demedetskaya

Point / Vector



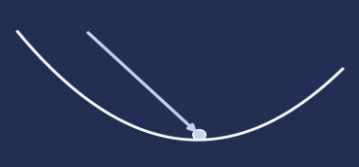
$x=(3,2,2)$

Plane / Separator



$3x_1+x_2-1=0$

Loss / Optimization



$\theta \leftarrow \theta-\gamma\nabla L(\theta)$

Classifier

$h(x)\in\{-1,+1\}$

Geometry gives us representation

Optimization gives us learning

Classification gives us prediction

Foundations for early machine learning concepts and linear classification

Part I — Framing, Notation, and Roadmap

How the course’s mathematical language becomes machine learning language

vector — hyperplane — classifier

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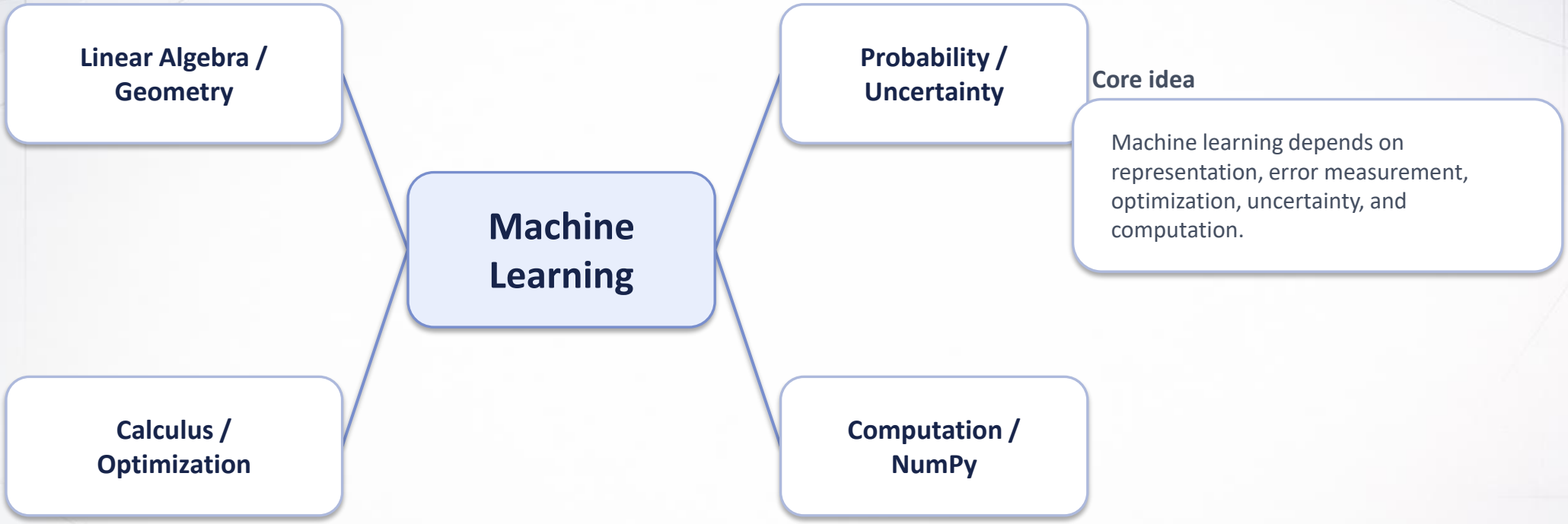
Why These Foundations Come First

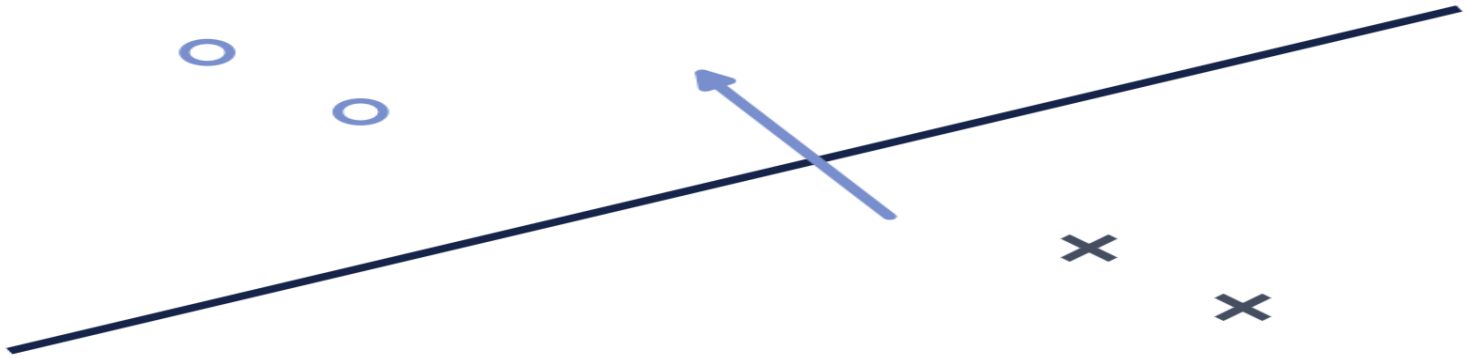


Core idea

Machine learning depends on representation, error measurement, optimization, uncertainty, and computation.

Representation, error, improvement, uncertainty, and computation all arrive before learning can happen.

Geometry Gives Us Representation



Geometry lets us place examples in space, compare directions, and define boundaries.

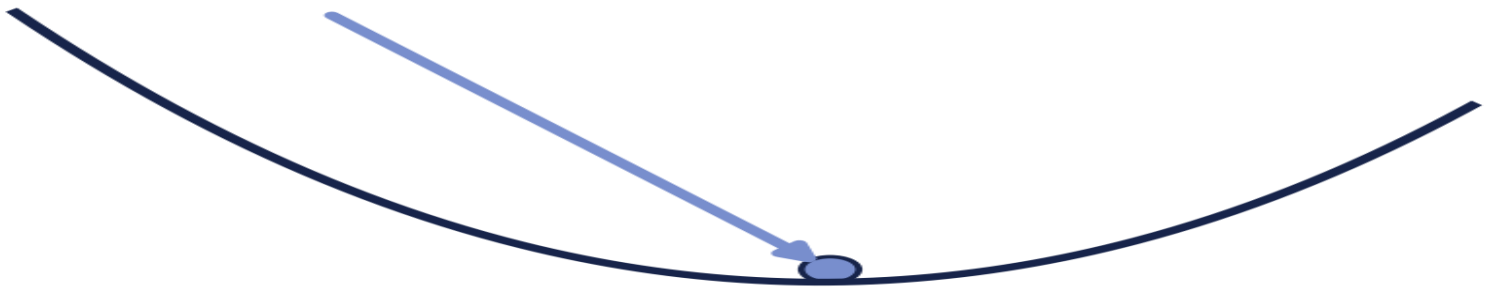
vectors represent objects • space gives structure • boundaries separate regions

$x \in R^d$ • $\theta \cdot x + \theta_0 = 0$

point/vector $x=(3,2,2)$ and boundary $3x_1+x_2-1=0$

Without geometry, there is no meaningful space in which learning can happen.

Optimization Improves Parameters



Optimization gives us a rule for improving parameters by moving toward lower loss.

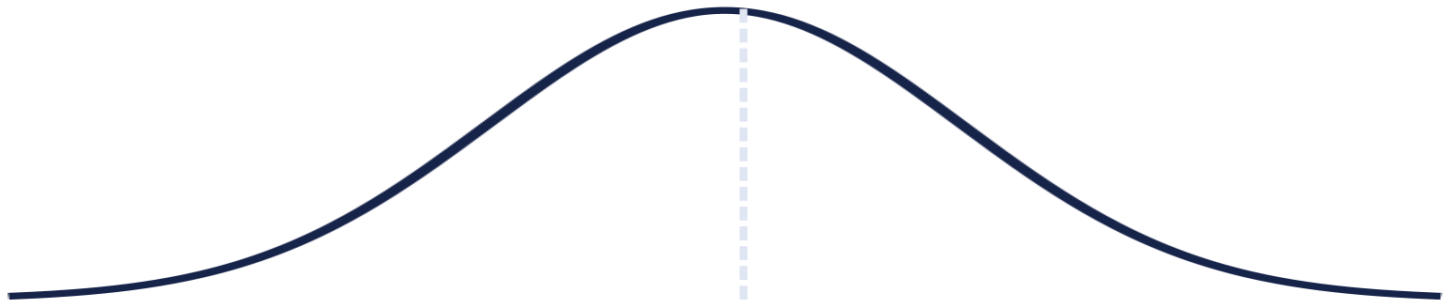
error becomes a function • lower is better • updates move downhill

$L(\theta)$ • $\theta \leftarrow \theta - \gamma \nabla L(\theta)$

Use a visible bowl surface or 1D loss curve.

Optimization is how a model changes from a bad fit into a better fit.

Probability Models Uncertainty



Probability lets us describe uncertainty, variability, and noisy data.

outcomes can vary • uncertainty can be modeled • ML must reason under uncertainty

$$X \sim N(\mu, \sigma^2)$$

Use one Gaussian with clear mean and spread.

Real data is not perfectly clean, so machine learning must reason probabilistically.

Computation Makes the Math Executable

```
[1,  0,  1]
[0,  1,  1]
```

Computation turns the math into something we can evaluate, repeat, and optimize.

vectors become arrays • formulas become code • repeated updates become algorithms

$$A\,x$$

Use a clear matrix-vector visual or array layout.

Without computation, the formulas stay on paper.

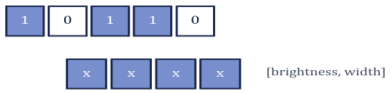
The Big Story: From Data to Decisions

Machine learning turns real-world objects into numerical representations, places them in a geometric space, and learns rules that predict labels or values.

Raw Object

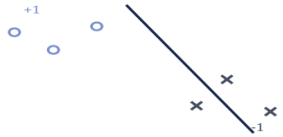
real-world input

Feature Vector



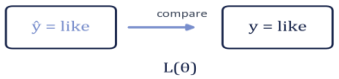
machine-readable x

Geometry / Model



space + boundary

Prediction

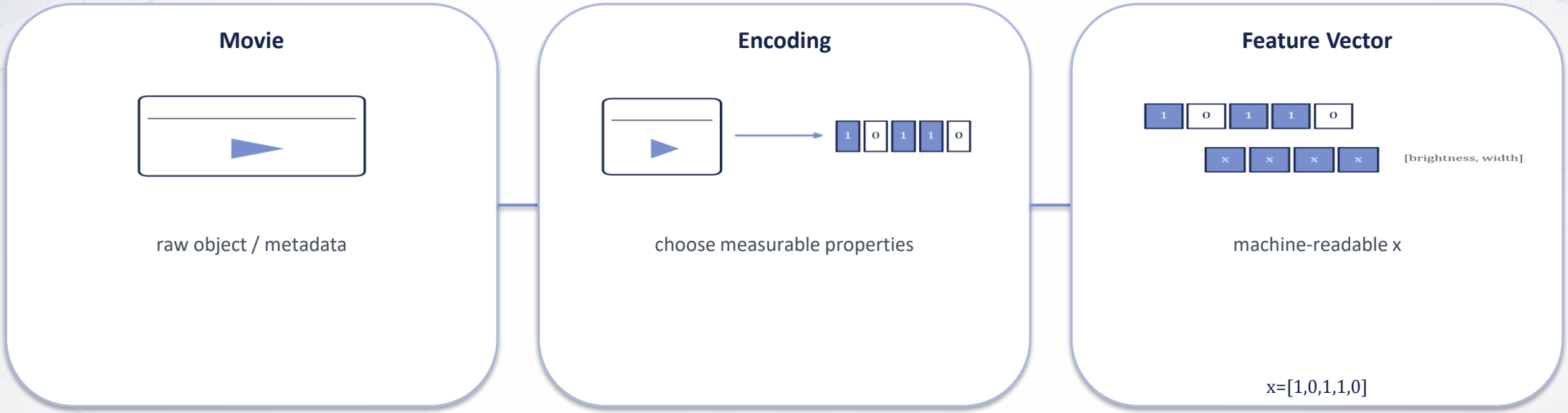


label / error

Representation comes first: objects become vectors, vectors enter geometry, and geometry supports prediction.

Example A — From Movie Object to Feature Vector

A real object becomes a machine-readable representation.



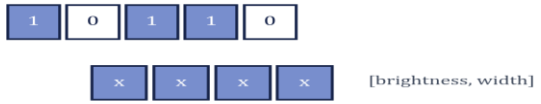
object • encoding • vector

Representation means converting a real object into a vector that a model can use.

Example A — From Feature Vector to Prediction

Once represented numerically, the object can enter a geometric model and produce a prediction.

Vector x



input to the model

Geometric Space

point in feature space

Prediction

like / dislike

$h(x)=\text{sign}(\theta \cdot x+\theta_0)$

vector • space • prediction

A prediction is made only after the object has been turned into a vector.

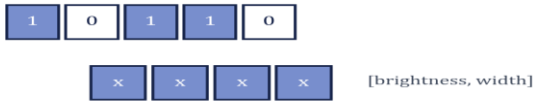
Example B — From Digit Image to Features

A handwritten image can also be converted into a numerical representation.

Digit Image

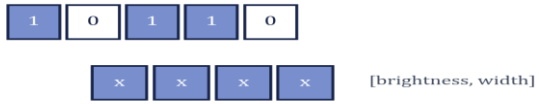
raw handwritten input

Measured Features



brightness / width

Vector x



numeric representation

x=[brightness,width]

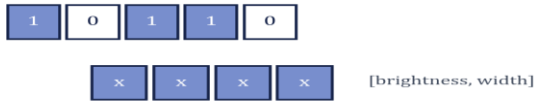
image • measurement • vector

The same representation idea works for images too.

Example B — From Features to Class Label

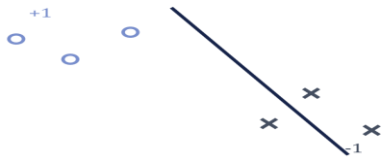
Once represented numerically, the image can be placed in space and classified.

Vector x



numeric input to the model

Class Regions



boundary / regions in space

Output Label

predicted class

$h(x)=7$

vector • boundary • class

A class prediction is made after the image becomes a vector in a geometric space.

Running Examples We Will Reuse — Geometry and Optimization

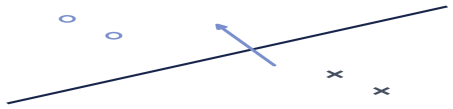
These examples will return throughout the deck as anchors.

Point / Vector

$x=(3,2,2)$

geometry anchor

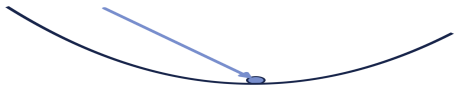
Plane / Hyperplane



$3x_1+x_2-1=0$

boundary anchor

1D Optimization



$f(x)=\frac{1}{3}x^3-x^2-3x+10$

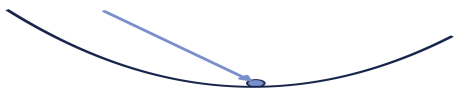
optimization anchor

geometry and optimization examples we will reuse often

Running Examples We Will Reuse — Probability, Computation, and Supervised Learning

These examples will return throughout the deck as anchors.

Gradient



$f(\theta)=\theta_1^2+\theta_2^2$

descent anchor

Gaussian

$X \sim N(1,2)$

probability anchor

Movie Recommender

5-feature vector + label ±1

supervised-learning anchor

probability, computation, and supervised-learning anchors

Notation You Must Read Fluently — Objects and Coordinates

In machine learning, the same letters are reused across different mathematical objects. Context matters.

symbol	meaning	visual example
$\mathbf{x}$	point / feature vector / example	$x=(3,2,2)$
$\mathbf{x_i}$	i-th coordinate	$x_2=2$
$\mathbf{A}$	matrix	A x
$\mathbf{y}$	label	$y=+1$

$x \in R^d$ • $A \in R^{m \times n}$ • $y \in \{-1,+1\}$

Notation You Must Read Fluently — Parameters, Classifiers, and Loss

These symbols will appear repeatedly in linear models, classification, and optimization.

symbol	meaning	visual example
θ	parameter vector / normal vector	$\theta \cdot x + \theta_0 = 0$
θ_0	offset / intercept	boundary shift
h	classifier / prediction rule	$h(x) = \mathrm{sign}(\theta \cdot x + \theta_0)$
L	loss	$L(\theta)$

$\theta \in R^d$ • $\theta \cdot x + \theta_0 = 0$ • $h(x) = \mathrm{sign}(\theta \cdot x + \theta_0)$ • $L(\theta)$

Point, Vector, Feature Vector: Same Shape, Different Role

Point

location in coordinate space

$x=(3,2,2)$

Vector

arrow from the origin

$x \in R^3$

Feature Vector

encoded example for prediction

$x \mapsto$ feature vector

location in space • displacement from origin • encoded example for prediction

$x=(3,2,2)$ • $x \in R^3$ • $x \mapsto$ feature vector

One Object, Three Interpretations

The coordinates did not change. The interpretation changed.

x=(3,2,2)

Point

same coordinates, location in space

same coordinates

Vector

same coordinates, displacement from origin

different role

Feature Vector

same shape, encoded movie example

context changes meaning

x=(3,2,2) • movie → x

The shape of the mathematical object stays the same; only the interpretation changes.